



Customer Shopping Behavior Analysis

End-to-end data analytics project – from raw data to business insights

PYTHON · POSTGRESQL · POWER BI



Project Overview

Goal

Analyze customer purchasing patterns, segmentation, and revenue trends

Scope

Subscription impact, customer behavior, and product performance

Output

Interactive Power BI dashboard with actionable business insights

Tools & Technologies



Python

Data cleaning & feature engineering



Power BI

Interactive visualization & dashboards



PostgreSQL

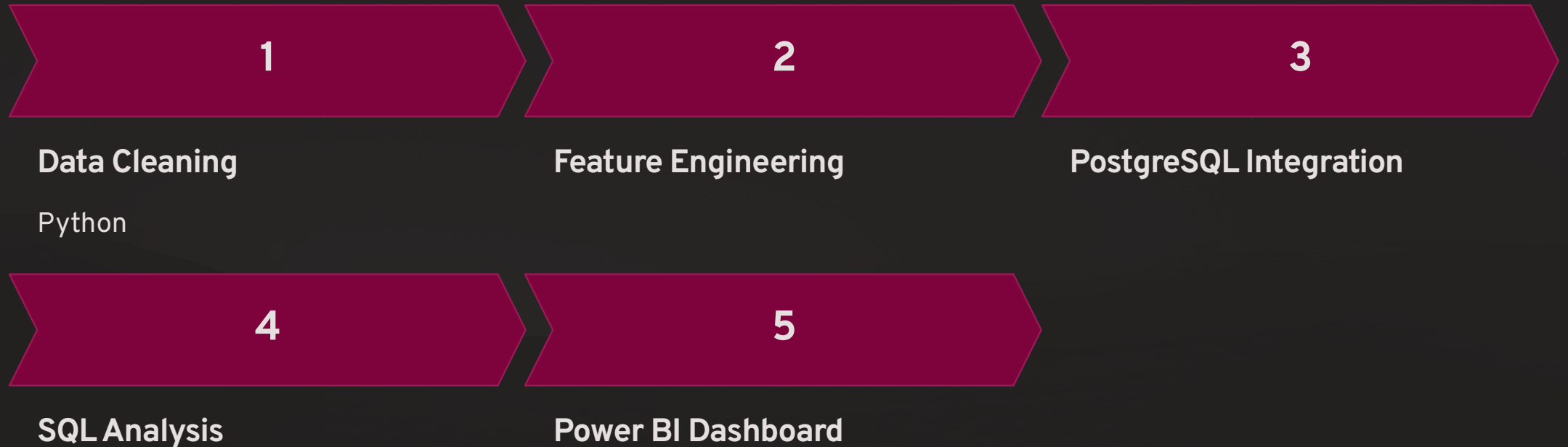
SQL business analysis & queries



Jupyter & Pandas

Exploratory data analysis

Project Workflow



Five-stage pipeline transforming raw data into interactive business intelligence.

DATA PREPARATION

Data Cleaning & Feature Engineering



- Missing review ratings → **category-wise median**
- Standardized column names
- Created **age_group** column
- Purchase frequency → numerical days
- Removed redundant columns

SQL Business Questions

Revenue by Gender

High Spenders & Discounts

Top-Rated Products

Subscription Spending

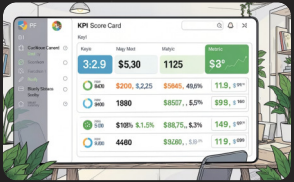
Customer Segmentation

Top Products by Category

Repeat Buyer Subscriptions

Revenue by Age Group

Power BI Dashboard Features



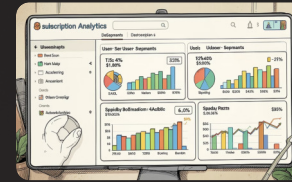
KPI Cards

Key metrics at a glance



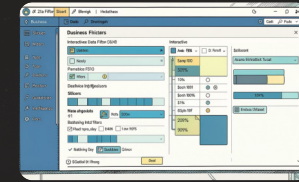
Revenue & Category Sales

Trends and breakdowns



Subscription Analysis

Subscriber vs non-subscriber behavior



Interactive Filters

Age groups, categories & slicers

Key Insights



Clothing Leads

Highest revenue category



Middle-Aged Buyers

Top contributors to sales



Subscribers Spend More

Higher average order value



Repeat → Subscribe

Repeat buyers more likely to subscribe



Discounts Drive Sales

Influenced purchases for specific products

Key Metrics at a Glance

5

Workflow Stages

End-to-end pipeline

8

SQL Questions

Business analysis queries

5

Dashboard Features

Interactive Power BI views

5

Key Insights

Actionable findings

Conclusion

This project demonstrates **end-to-end data analytics** – from raw data to business intelligence.



Data Cleaning



SQL Querying



Customer Segmentation



Dashboard Design



Business Insights